Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.75)

Metric

$$ds^2 = -\left(ar^2 + bt\right)^{\frac{2}{3}} \mathrm{d}t \otimes \mathrm{d}t + \left(ar^2 + bt\right)^{\frac{2}{3}} \mathrm{d}r \otimes \mathrm{d}r + \left(ar^2 + bt\right)^{\frac{2}{3}} r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + \left(ar^2 + bt\right)^{\frac{2}{3}} r^2 \sin\left(\theta\right)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$A = \left(ar^2 + bt\right)^{\frac{2}{3}}$$

Null Tetrad

Covariant components

$$l = \left(\left(ar^2 + bt\right)^{\frac{2}{3}}\right) dt + \left(\left(ar^2 + bt\right)^{\frac{2}{3}}\right) dr$$

$$n = \left(\frac{1}{2}\right) dt + \left(-\frac{1}{2}\right) dr$$

$$m = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r\right) d\theta + \left(\frac{1}{2} i\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r \sin\left(\theta\right)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r\right) d\theta + \left(-\frac{1}{2} i\sqrt{2}\left(ar^2 + bt\right)^{\frac{1}{3}} r \sin\left(\theta\right)\right) d\phi$$

Contravariant components

$$l = (-1)\ dt + (1)\ dr$$

$$n = \left(-\frac{1}{2\,(ar^2+bt)^{\frac{2}{3}}}\right)\ dt + \left(-\frac{1}{2\,(ar^2+bt)^{\frac{2}{3}}}\right)\ dr$$

$$m = \left(\frac{\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r}\right)\ d\theta + \left(\frac{i\,\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r\sin(\theta)}\right)\ d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r}\right)\ d\theta + \left(-\frac{i\,\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r\sin(\theta)}\right)\ d\phi$$

Directional Derivatives

$$D = -\partial_t + \partial_r$$

$$\Delta = -\frac{1}{2\,(ar^2+bt)^{\frac{2}{3}}}\,\partial_t - \frac{1}{2\,(ar^2+bt)^{\frac{2}{3}}}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r\sin(\theta)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\,(ar^2+bt)^{\frac{1}{3}}\,r\sin(\theta)}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{5\,ar^2}{3\,(ar^3 + brt)} + \frac{br}{3\,(ar^3 + brt)} - \frac{bt}{ar^3 + brt}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{5\,ar^2}{6\,(ar^3 + brt)(ar^2 + bt)^{\frac{2}{3}}} - \frac{br}{6\,(ar^3 + brt)(ar^2 + bt)^{\frac{2}{3}}} - \frac{bt}{2\,(ar^3 + brt)(ar^2 + bt)^{\frac{2}{3}}}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos(\theta)}{4\,(ar^2 + bt)^{\frac{1}{3}} r \sin(\theta)}$$

$$\beta = \frac{\sqrt{2}\cos(\theta)}{4\,(ar^2 + bt)^{\frac{1}{3}} r \sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{2\,ar}{3\,(ar^2 + bt)} - \frac{b}{3\,(ar^2 + bt)}$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{10\,a^2r^2}{9\,(a^2r^4 + 2\,abr^2t + b^2t^2)} - \frac{16\,abr}{9\,(a^2r^4 + 2\,abr^2t + b^2t^2)} - \frac{2\,abt}{3\,(a^2r^4 + 2\,abr^2t + b^2t^2)} + \frac{4\,b^2}{9\,(a^2r^4 + 2\,abr^2t + b^2t^2)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{11\,a^2r^2}{18\,(a^2r^4 + 2\,abr^2t + b^2t^2)(ar^2 + bt)^{\frac{2}{3}}} - \frac{abt}{6\,(a^2r^4 + 2\,abr^2t + b^2t^2)(ar^2 + bt)^{\frac{2}{3}}} + \frac{b^2}{9\,(a^2r^4 + 2\,abr^2t + b^2t^2)(ar^2 + bt)^{\frac{2}{3}}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = {} & \frac{5\,a^2r^2}{18\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}} + \frac{4\,abr}{9\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}} \\ & - \frac{abt}{6\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}} + \frac{b^2}{9\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)(ar^2 + bt)^{\frac{1}{3}}}\end{aligned}$$

$$\Lambda = -\frac{5\left(ar^2 + bt\right)^{\frac{1}{3}}a^2r^2}{18\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)} - \frac{\left(ar^2 + bt\right)^{\frac{1}{3}}abt}{2\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)} - \frac{\left(ar^2 + bt\right)^{\frac{1}{3}}b^2}{18\,(a^3r^6 + 3\,a^2br^4t + 3\,ab^2r^2t^2 + b^3t^3)}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$\Psi_2 = 0$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"